

Technology Stack

Date	4 july 2026
Team ID	xxxxxx
Project Name	Personalized network assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js + HTML/CSS + JavaScript	React.js is chosen for creating an interactive and responsive user interface. It supports reusable components and provides a smooth experience for users interacting with the personalized network assistant.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (Django/Flask)	Python is used for backend development because it provides fast processing, easy integration with AI/ML models, and efficient handling of user requests and network operations.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MongoDB	MongoDB is selected to store user profiles, preferences, chat history, and network data. It handles flexible and unstructured data efficiently
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	AWS (Amazon Web Services)	AWS provides scalable, reliable, and cost-effective hosting for deploying the personalized network assistant and managing server resources.
5	Version Control & CI/CD	GitHub + GitHub Actions	GitHub enables team collaboration and source code management, while GitHub Actions helps automate testing and deployment processes.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Google Speech-to-Text API, OpenAI API, Network Monitoring APIs	These APIs provide voice interaction, intelligent responses, and real-time network monitoring features to improve the assistant's capabilities